Trigonometric Identities and Equations — Illegal Cancellation and Lost Solutions

Answer Key

Precalculus

Quick Answers

1. 0, 60, 180, 300
2. $\cos(x) + 1$, 1.5
3. 60, 90, 270, 300
4. 2.5
5. 30, 150, 210, 330
6. 90, 180
7. 0, 120, 240
8. —

Curriculum

Level: Precalculus
HSF-TF.A.2, HSF-TF.B.5 – *problem 6*
HSF-TF.C.8 – *problems 1, 2, 3, 4, 5, 7, 8*
Difficulty 1–5 of 5 across 9 tagged checks.

Common wrong answers

2. *If they got 0.5: cancelled the squared cosine against the single cosine term by term, which leaves the cosine itself and drops the added one*
4. *If they got 1.75: cancelled the cosine in the denominator against the cosine in the numerator even though that cosine is one term of a sum, not a factor*

1. Find and fix: $2\sin\theta\cos\theta = \sin\theta$.

(a) Step 1: name the slip. Dividing by $\sin\theta$ silently assumes $\sin\theta \neq 0$. Every angle where $\sin\theta = 0$ is thrown out of the problem before it is ever tested — and those angles satisfy the original equation, since both sides become 0.
Step 2: move everything to one side and factor instead: $2\sin\theta\cos\theta - \sin\theta = 0$, so $\sin\theta\,(2\cos\theta - 1) = 0$.
(b) Step 3: the zero-product property keeps both branches. $\sin\theta = 0$ gives $\theta = 0°, 180°$; $\cos\theta = \frac{1}{2}$ gives $\theta = 60°, 300°$.
Step 4:

$$\boxed{\theta = 0°,\ 60°,\ 180°,\ 300°}$$

Step 5: check the two Rafi lost — at $\theta = 0°$ both sides are 0 ✓, and at $\theta = 180°$ both sides are 0 ✓.
Teaching point: a solution set that is a strict subset of the true one is the signature of a division. Ask what was divided by and set it to zero.

2. Find and fix: $\dfrac{\cos^2 x - 1}{\cos x - 1}$**.**

(a) Step 1: name the slip. Bea cancelled term by term. Cancelling is division, and division distributes over *factors*, never over the separate terms of a sum or difference.
Step 2: factor the numerator as a difference of squares: $\cos^2 x - 1 = (\cos x - 1)(\cos x + 1)$. Now $\cos x - 1$ is a genuine common factor, so it may be cancelled: $\dfrac{(\cos x - 1)(\cos x + 1)}{\cos x - 1} = \boxed{\cos x + 1}$ (valid for $\cos x \neq 1$).

(b) Step 3: test at a value with $\cos x = 0.5$. The original gives $\dfrac{0.25 - 1}{0.5 - 1} = \dfrac{-0.75}{-0.5} = \boxed{1.5}$, while Bea's $\cos x$ gives 0.5 — not equal, so her form is wrong.
Step 4: check the correct form at the same value: $\cos x + 1 = 0.5 + 1 = 1.5$ ✓, matching the original exactly.
Watch for: 0.5. One numerical test is enough to kill a wrong simplification, and it takes ten seconds.

3. Find and fix: $2\cos^2\theta = \cos\theta$**.**

(a) Step 1: never divide by $\cos\theta$. Move everything to one side: $2\cos^2\theta - \cos\theta = 0$, then factor: $\cos\theta\,(2\cos\theta - 1) = 0$.

(b) Step 2: solve each branch. $\cos\theta = 0$ gives $\theta = 90^\circ, 270^\circ$; $\cos\theta = \frac{1}{2}$ gives $\theta = 60^\circ, 300^\circ$.
Step 3:

$$\boxed{\theta = 60^\circ,\ 90^\circ,\ 270^\circ,\ 300^\circ}$$

Step 4: Nils's division discarded exactly the branch $\cos\theta = 0$, so he lost 90° and 270°. Check one: at $\theta = 90^\circ$, $2(0)^2 = 0 = \cos 90^\circ$ ✓.
Teaching point: the lost solutions are always the zeros of whatever you divided by — here, of $\cos\theta$. That makes them easy to recover once the error is named.

4. Find and fix: $\dfrac{\sin^2 x + \cos x}{\cos x}$**.**

(a) Step 1: the numerator is a *sum*, and $\cos x$ is only one of its two terms. A denominator divides the whole numerator, so it can only be cancelled against a common factor of *every* term. Here $\sin^2 x$ has no $\cos x$ in it, so nothing cancels.
Step 2: split the fraction instead, which is legal: $\dfrac{\sin^2 x}{\cos x} + \dfrac{\cos x}{\cos x} = \dfrac{\sin^2 x}{\cos x} + 1$.

(b) Step 3: test where $\sin^2 x = 0.75$ and $\cos x = 0.5$: $\dfrac{0.75}{0.5} + 1 = 1.5 + 1 = \boxed{2.5}$, while the student's $\sin^2 x + 1$ gives 1.75.
Step 4: check against the original fraction directly: $\dfrac{0.75 + 0.5}{0.5} = \dfrac{1.25}{0.5} = 2.5$ ✓.
Watch for: 1.75. The gap of 0.75 is exactly the term that was wrongly left undivided.

5. Find and fix: $4\sin^2\theta = 1$**.**
(a) Step 1: name the slip. $\sqrt{u^2} = |u|$, so taking a square root gives $|2\sin\theta| = 1$, i.e. $\sin\theta = \pm\frac{1}{2}$. Tam kept only the positive branch and lost the negative one.
Step 2: safer still, factor as a difference of squares: $4\sin^2\theta - 1 = (2\sin\theta - 1)(2\sin\theta + 1) = 0$, which shows both branches at once and never involves a square root.
(b) Step 3: $\sin\theta = \frac{1}{2}$ gives $\theta = 30°, 150°$; $\sin\theta = -\frac{1}{2}$ gives $\theta = 210°, 330°$.
Step 4:

$$\boxed{\theta = 30°,\ 150°,\ 210°,\ 330°}$$

Step 5: check one from the lost branch — at $\theta = 210°$, $\sin\theta = -\frac{1}{2}$ and $4\left(\frac{1}{4}\right) = 1$ ✓.
Teaching point: squaring destroys sign information, so undoing it must restore both signs. Factoring never loses them in the first place.

6. Find and fix: $\sin\theta - \cos\theta = 1$**.**
(a) Step 1: name the slip. Squaring is not reversible: if $A = B$ then $A^2 = B^2$, but $A^2 = B^2$ only forces $A = \pm B$. So Ola's squared equation is satisfied by the solutions of $\sin\theta - \cos\theta = 1$ *and* by those of $\sin\theta - \cos\theta = -1$. Squaring added solutions rather than losing them — which is why every candidate has to be tested in the original.
(b) Step 2: test all four candidates in $\sin\theta - \cos\theta = 1$. At 0°: $0 - 1 = -1$ ×. At 90°: $1 - 0 = 1$ ✓. At 180°: $0 - (-1) = 1$ ✓. At 270°: $-1 - 0 = -1$ ×.
Step 3:

$$\boxed{\theta = 90°,\ 180°}$$

Step 4: the two rejected angles satisfy $\sin\theta - \cos\theta = -1$, exactly the extra equation squaring smuggled in ✓.
Teaching point: dividing removes solutions, squaring adds them. Both are fixed by the same discipline — factor rather than divide, and substitute every candidate back.

7. Find and fix: $\cos 2\theta = \cos\theta$**.**
(a) Step 1: name the slip. "Cancelling the cosine" treats cos as a multiplier, but it is a function, and it is not one-to-one on a full turn: $\cos 120° = \cos 240°$ while $120° \neq 240°$. Equal outputs therefore do not force equal inputs, and Yuki's step keeps only the solutions where the angles happen to match.
(b) Step 2: use $\cos 2\theta = 2\cos^2\theta - 1$ and move everything to one side: $2\cos^2\theta - \cos\theta - 1 = 0$.
Step 3: factor: $(2\cos\theta + 1)(\cos\theta - 1) = 0$, so $\cos\theta = -\frac{1}{2}$ or $\cos\theta = 1$.
Step 4: $\cos\theta = -\frac{1}{2}$ gives $\theta = 120°, 240°$, and $\cos\theta = 1$ gives $\theta = 0°$.
Step 5:

$$\boxed{\theta = 0°,\ 120°,\ 240°}$$

Step 6: check the two Yuki lost — at $\theta = 120°$, $\cos 240° = -\frac{1}{2} = \cos 120°$ ✓; same at 240°, where $\cos 480° = \cos 120° = -\frac{1}{2}$ ✓.
Teaching point: the only cancelling that is ever legal is cancelling a common *factor*. Applying an inverse function to both sides is a different move with its own conditions.

8. Explain: what division removes and squaring adds.

Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.

(a) Dividing both sides by an expression E replaces the original equation with one that is only equivalent *where* $E \neq 0$. The step silently assumes E is never zero, so every angle making $E = 0$ is deleted from the problem before it can be tested — and those angles often do solve the original, because they make both sides zero at once. The repair is structural: move everything to one side and factor, so the zero-product property reports each branch, including $E = 0$.

(b) Cancelling is division, and division distributes over factors, not over terms. In $\frac{ab}{a}$ the a divides the whole numerator, so it cancels; in $\frac{a+b}{a}$ the a is only part of what is being divided, so cancelling it divides one term and not the other. A number makes it plain: $\frac{2+6}{2} = 4$, but "cancelling the 2" would give 6. The correct move is to split the fraction: $\frac{2}{2} + \frac{6}{2} = 1 + 3 = 4$.

(c) Squaring is not reversible: $A = B$ implies $A^2 = B^2$, but $A^2 = B^2$ only implies $A = \pm B$. The squared equation therefore carries the solutions of $A = -B$ as well, so every candidate must be substituted into the original to see which equation it actually solved. Squaring *adds* solutions (extraneous ones), while dividing *removes* them (lost ones). The asymmetry is that squaring widens the solution set and can be repaired afterwards by checking, whereas dividing narrows it and cannot be repaired afterwards — the lost solutions are simply gone unless you go back and recover the zeros of the divisor.

What a full-credit answer must contain: the non-zero assumption behind division in (a), the factor-versus-term distinction with a concrete numerical example in (b), and in (c) the $A = \pm B$ argument together with the correct direction of each effect — squaring adds, dividing removes.